LESSON 9.6
DESCRIBING RIGID TRANSFORMATIONS

LESSON INTRODUCTION

MA.8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.

MA.8.GR.2.3 Describe and apply the effect of a single transformation on two-dimensional figures using coordinates and the coordinate plane.

LEARNING TARGETS

- I can describe the effect of a single transformation.
- I can describe the effect of a transformation on the coordinate plane.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation and symmetries of translation of a geometric figure.

ESSENTIAL QUESTIONS

- Describe the effect of the transformation.
- What are the characteristics of translations, rotations, and reflections?
- How are translations, rotations, and reflections similar? How are they different?

LESSON NARRATIVE

Students practice describing rigid transformations on the coordinate plane. Students discuss how to match diagrams of transformations to their descriptions. Students also independently identify and describe transformations.

COMPONENT SUMMARY

9.6.1 WARM-UP *(~5 MIN)*

Solve a number puzzle

9.6.2 COLLABORATION *(20 MIN)*

Match transformations to their descriptions

9.6.3 GUIDED PRACTICE *(10 MIN)*

Describe transformations

9.6.4 YOUR TURN! *(5 MIN)*

Practice describing transformations

9.6.5 WRAP-UP *(~5 MIN)*

Determine which transformation was performed

RESOURCES

GLOSSARY

Key Terms:

- Translation
- Rotation
- Reflection

Additional Terms: N/A

MATERIALS

- Card Sort
- (Optional) Graph Paper
- (Optional) Cutouts of Figures
- (Optional) Print outs of puzzle template from Warm-Up

ADDITIONAL PREPARATION

- Enlarged visual representations of figures could be created on card stock and cut out prior to lesson.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not notice changes in the orientation of the figures and may therefore confuse rotations and reflections.

THINGS TO CONSIDER

- Consider having students perform a transformation by determining the location of one point at a time. Have students physically perform transformations using cutouts of figures or to use patty paper to reinforce each transformation.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

During 9.6.2 Collaboration, circulate, collect, and make a visual display of the vocabulary, phrases, and representations students use to describe each transformation. During the debrief, make connections between how similar ideas might be communicated and represented in different ways. Make note of common words or phrases that you hear and display this list for students to refer to throughout the lesson.

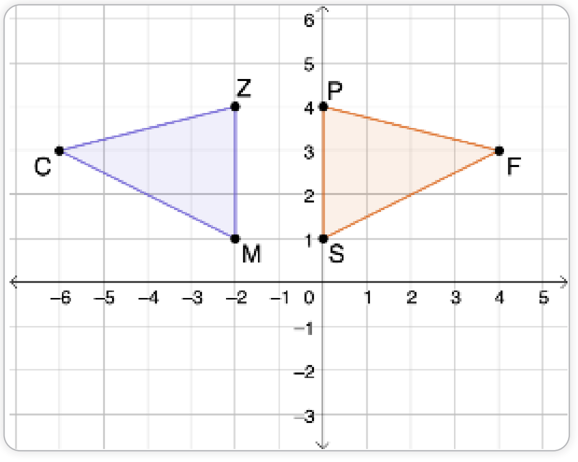
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 9.6.4 Your Turn! students independently describe transformations. Prompt students to look back at the summary table in 9.6.2 Collaboration and descriptions written in 9.6.3 Guided Practice. Ask students to consider the patterns and structures they notice about each transformation.

DIFFERENTIATE INSTRUCTION

The images of the transformations provide a visual entry-point in 9.6.4 Try It! Discussions about the descriptions of the transformations during 9.6.2 Collaboration provide auditory entry-point. Offer a kinesthetic entry-point by prompting students to manipulate cutouts of the figures as they explore and describe transformations.



9.6.1 WARM-UP: EIGHT-DIGIT PUZZLE

TEACHER MOVES

Students may work individually or with a partner. Consider printing out several templates of the puzzle so students can save their attempts. Encourage students to discuss the process they used to solve the puzzle. Prompt students to save each attempt (not erase them). Ask students how their movements of numbers between their attempts could be described as transformations.

QUESTIONING

Challenge students to create a number puzzle. Then, students can switch puzzles with a partner and solve each other's puzzles.

9.6.1 Warm-Up: Eight-Digit Puzzle

1. Place the numbers 1 through 8 in the rectangles below so that no two consecutive numbers are next to each other horizontally, vertically, or diagonally.

	3	5	
7	1	8	2
	4	6	

9.6.2 Collaboration: Transformation Matching Cards

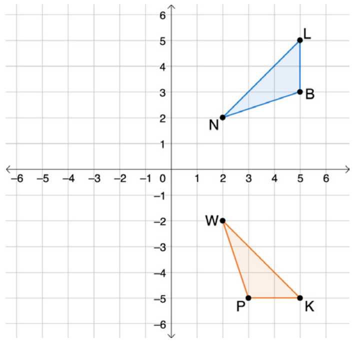
1. Work with your partner to match the transformation cards with the descriptions of the transformations. Record your matches in the table using the letter shown on the cards.

Transformation Card	Description Card
1	*D*
2	*F*
3	*G*
4	*A*
5	*E*
6	*B*
7	*H*
8	*C*

2. Write a brief summary of what you noticed about the location and orientation of the image in comparison to the preimage for each type of transformation.

Transformation	Location	Orientation
Translation	*The location changes.*	*The orientation stays the same.*
Reflection	*The location changes.*	*The orientation changes – the image is a mirror of the preimage.*
Rotation	*The location changes.*	*The orientation changes, but the orientation depends on the degrees of rotation.*

3. Nigel and Elena are discussing what single transformation took place to map ΔBNL to ΔPWK.



a. Nigel claims that the transformation is a rotation about the origin. Elena says the transformation is a reflection. Which student do you agree with?
I agree with Nigel because the orientation of the image is not right for a reflection.

b. If you agree with Nigel, what is the angle of rotation and direction? If you agree with Elena, what is the line of reflection?
ΔBNL was rotated $90°$ clockwise.
OR
ΔBNL was rotated $270°$ counterclockwise.

9.6.2 COLLABORATION: TRANSFORMATION MATCHING CARDS

TEACHER MOVES

Encourage students to pay attention to the way the transformations are described on the cards as they can reference these descriptions and use them as a model for the descriptions. They are asked to compose in 9.6.3 Guided Practice and 9.6.4 Your Turn!

QUESTIONING

If students finish the matching activity quickly, prompt them to write a justification for each match. How are they certain that their matches are correct?

ENRICHMENT

What similarities and differences do you notice? What do you notice about how a reflection, translation, and rotation are each described?

TEACHER MOVES

Collect and Display

Circulate, collect, and make a visual display of the vocabulary, phrases, and representations students use to describe each transformation. During the debrief, make connections between how similar ideas might be communicated and represented in different ways. Make note of common words or phrases that you hear and display this list for students to refer to throughout the lesson.

DIFFERENTIATE INSTRUCTION

Provide graph paper and cutouts of figures or patty paper to allow students to physically perform the transformations. Encourage students to describe the transformations verbally.

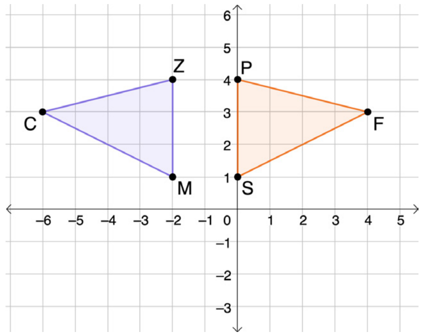
9.6.3 GUIDED PRACTICE: DESCRIBING TRANSFORMATIONS

TEACHER MOVES

How can you check to confirm that this is a reflection? What do you notice about the corresponding points? How do the coordinates of C and F compare? What about Z and P? What about M and S? What would the coordinates of the image be if it were reflected over the x-axis?

9.6.3 Guided Practice: Describing Transformations

1. On the coordinate grid shown, ΔZCM is the preimage, and ΔPFS is the image after a single transformation.



Select the statement that best describes the transformation. Then, complete the chosen statement.

- ☐ ΔZCM can be mapped to ΔPFS by translating it _____ units to the right.
- ☐ ΔZCM can be mapped to ΔPFS by reflecting it over line $x = -1$.
- ☐ ΔZCM can be mapped to ΔPFS by rotating it ______° clockwise about the origin.

TEACHER MOVES

Students may write the name of the transformation without providing details such as the line of reflection, the direction and angle of rotation, or the distance and direction of translation. Prompt students to refer to the description cards from 9.6.2 Collaboration. Students should use these descriptions as a model for what to write in their descriptions.

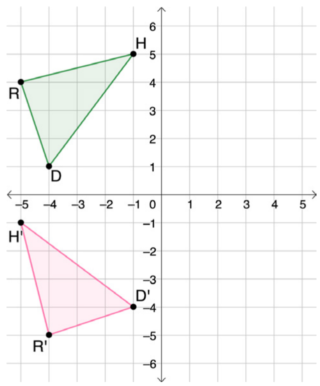
DIFFERENTIATE INSTRUCTION

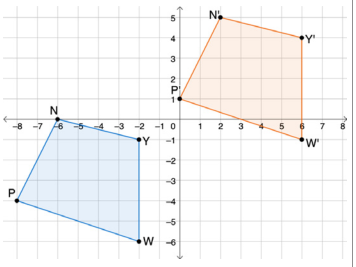
Prompt students to stand and face the front of the room. Label this 0°. Have students turn a quarter turn either way. Discuss that a quarter turn is 90°. Ask students what the angle of rotation would be if they made a half turn and a three-quarter turn. Ask if the measure depends on the direction they turned (rotated).

When describing a transformation on the coordinate plane, the description should include the type of transformation as well as the following information.

- **Translation**: the direction and number of units moved
- **Rotation** about the origin: the angle and direction of the rotation
- **Reflection**: the line of reflection

2. Describe each transformation shown.

Transformation	Description
ΔHDR is the preimage and $\Delta H'D'R'$ is the image. 	*ΔHDR can be mapped to $\Delta H'D'R'$ by a rotation of 90° counterclockwise about the origin.* *OR* *ΔHDR can be mapped to $\Delta H'D'R'$ by a rotation of 270° clockwise about the origin.*

Transformation	Description
Quadrilateral $NYWP$ is the preimage and quadrilateral $N'Y'W'P'$ is the image.  	*Quadrilateral $NYWP$ can be mapped to quadrilateral $N'Y'W'P'$ by a translation of 5 units up and 8 units to the right.* *OR* *Quadrilateral $NYWP$ can be mapped to quadrilateral $N'Y'W'P'$ by a translation of 8 units to the right and 5 units up.*

9.6.3 GUIDED PRACTICE: DESCRIBING TRANSFORMATIONS (CONTINUED)

ENRICHMENT

- What is the difference between the preimage and the image?
- How are the images similar? How are they different?
- Can you perform the transformation on graph paper based on the description you have written?

QUESTIONING

Students who finish early can create tessellations using their knowledge of transformations.

9.6.4 YOUR TURN!

ENRICHMENT

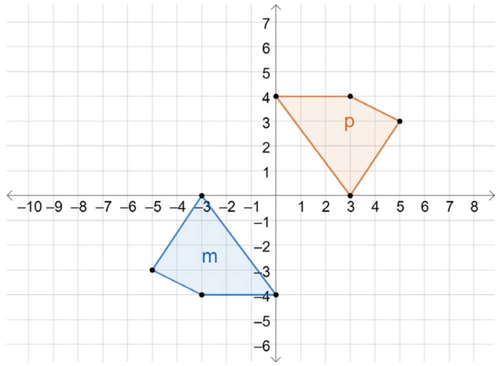
- How has the orientation changed?
- How have the corresponding points changed?
- What details are needed to perform this transformation?

DIFFERENTIATE INSTRUCTION

Provide sentence frames for students to use to describe the transformations.

9.6.4 Your Turn!

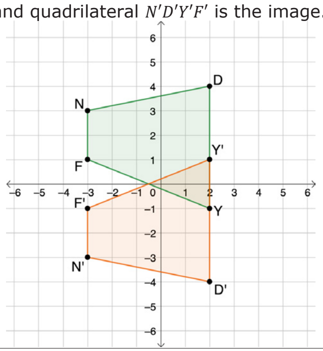
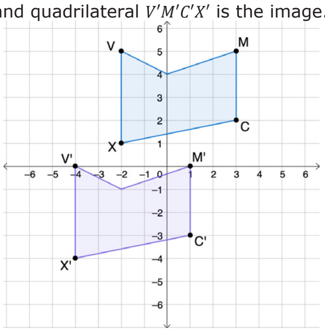
1. On the coordinate grid, figure p is the result of a transformation on figure m.



Select the statement that best describes the transformation. Then, complete the chosen statement.

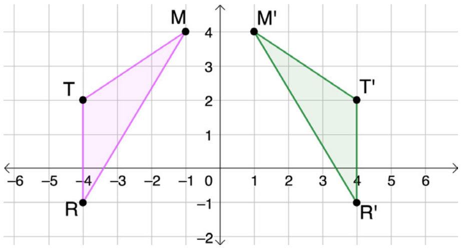
- ☐ Figure m was translated _____ units ○ right ○ left and _____ units ○ up ○ down to map to figure p.
- ☐ Figure m was reflected to map to figure p. The line of reflection is ________________.
- ☐ Figure m was rotated about the origin to map to figure p. The angle of rotation was <u>180°</u> counterclockwise.

2. Describe each transformation. Include the units of translation, line of reflection, or angle and direction of rotation as needed.

Transformation	Description
Quadrilateral $NDYF$ is the preimage and quadrilateral $N'D'Y'F'$ is the image. 	*Quadrilateral $NDYF$ can be mapped to quadrilateral $N'D'Y'F'$ by a reflection over the x-axis.*
Quadrilateral $VMCX$ is the preimage and quadrilateral $V'M'C'X'$ is the image. 	*Quadrilateral $VMCX$ can be mapped to quadrilateral $V'M'C'X'$ by a translation 2 units to the left and 5 units down.* *OR* *Quadrilateral $VMCX$ can be mapped to quadrilateral $V'M'C'X'$ by a translation 5 units down and 2 units to the left.*

9.6.5 Wrap-Up: Ticket Out the Door

1. ΔMTR, the preimage, and $\Delta M'T'R'$, the image, are shown on the coordinate plane.



Select and complete one statement to describe the transformation.

- ☐ To create the image, the preimage was translated ______ units right.
- ☐ To create the image, the preimage was reflected across the y-axis.
- ☐ To create the image, the preimage was rotated ______° clockwise about the origin.

9.6.5 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

If students finish early, ask them to perform the transformations described in the distractor options to the preimage.

